Question Bank and PYQs

Computer System

2 Mark Questions

Q1: What is a computer system? (2 Marks)

- A computer is an electronic device programmed to accept data as input, process it, and generate a result as output.
- A computer system consists of a computer along with additional hardware and software working together.
- It primarily comprises a central processing unit (CPU), memory, input/output devices, and storage devices.
- All these components function together as a single unit to deliver the desired output.

Q2: Define the Central Processing Unit (CPU). (2 Marks)

- The CPU is the electronic circuitry that carries out actual data processing.
- It is commonly referred to as the 'brain' of the computer.
- Physically, it can be placed on one or more microchips called integrated circuits (IC).
- The CPU fetches programs and data from memory to perform arithmetic and logic operations.

Q3: What are CPU registers? (2 Marks)

- Registers are local memory used by the CPU to store data and instructions during processing.
- They are part of the CPU chip itself and are limited in size and number.
- Different registers are used for storing data, instructions, or intermediate results.

Q4: Briefly explain the purpose of input devices. (2 Marks)

- Input devices are used to send control signals and data to a computer.
- They convert input data into a digital form that the computer system can accept.

- Examples include the keyboard, mouse, scanner, and touch screen.

Q5: What is the function of an output device? (2 Marks)

- An output device receives data from a computer system for display or physical production.
- It converts digital information back into a human-understandable form.
- Examples include monitors, speakers, and printers.

Q6: Define 'bit' and 'byte' in the context of computer memory. (2 Marks)

- A bit (binary digit) is the basic unit of memory and can be either 0 or 1.
- A byte consists of 8 bits.
- Computers use binary numbers to store and process all data.

Q7: What is cache memory? (2 Marks)

- Cache is a very high-speed memory placed between the CPU and primary memory.
- It speeds up CPU operations by storing copies of data from frequently accessed primary memory locations.
- It reduces the average time required to access data from the slower primary memory.

Q8: Distinguish between a 'soft-copy' and a 'hard-copy'. (2 Marks)

- A soft-copy refers to a document or image stored digitally on a hard disk or pen drive.
- A hard-copy is the physical form of the document or image after it has been printed.

Q9: What is a system utility? (2 Marks)

- System utilities are software used for the maintenance and configuration of a computer system.
- Some are shipped with the OS, such as disk defragmentation or formatting tools.
- Others improve performance, such as anti-virus software or disk cleaners.

Q10: Briefly explain the concept of a microcontroller. (2 Marks)

- A microcontroller is a small computing device with a CPU, fixed RAM, and ROM embedded on a single chip.
- It is designed for specific tasks, which reduces its size and cost.

- Common examples include those found in washing machines, digital cameras, and remote controllers.

3 Mark Questions

Q11: Explain the difference between RAM and ROM. (3 Marks)

- RAM (Random Access Memory) is volatile, meaning its contents are lost when power is turned off.
- RAM is used to store data temporarily while the computer is working.
- ROM (Read Only Memory) is non-volatile, meaning it retains its contents even without power.
- ROM stores small, permanent programs like the boot loader that starts the operating system.
- RAM allows both read and write operations, whereas ROM is primarily for reading.

Q12: Describe the three types of buses that make up the system bus. (3 Marks)

- The Data Bus is used to transfer data between different components and is bidirectional.
- The Address Bus transfers memory addresses from the CPU to main memory and is unidirectional.
- The Control Bus communicates control signals (like read or write) between components and is unidirectional.
- Collectively, these three buses form the System Bus.
- They act as physical wires for communication between the CPU, memory, and I/O devices.

Q13: What is the purpose of an Operating System? (3 Marks)

- The OS acts as a resource manager that manages hardware like the CPU, RAM, and disk.
- It provides services for building and running application programs.
- It provides a user interface (UI) to allow humans to interact with the computer.

- It handles system security and manages access for different users.
- It manages files, processes, and devices to ensure optimal system performance.

Q14: Explain the three main types of data: Structured, Semi-structured, and Unstructured. (3 Marks)

- Structured data follows a strict record structure and is usually organized in rows and columns, like a spreadsheet.
- Unstructured data does not follow a predefined format and includes things like audio, video, and social media posts.
- Semi-structured data has no rigid format but uses internal tags or markers to separate elements, like HTML or CSV files.
- Structured data is the easiest to sort and search.
- Unstructured data is often the most complex to capture and process.

Q15: Discuss the evolution of the microprocessor based on Moore's Law. (3 Marks)

- Moore's Law predicted that the number of transistors on a single chip would double every two years.
- This law also predicted that the cost of these chips would be halved over the same period.
- This led to Large Scale Integration (LSI) and later Very Large Scale Integration (VLSI).
- Modern chips use Super Large Scale Integration (SLSI) to fit millions of components on one IC.
- As a result, processing power has increased exponentially while the physical size of devices has decreased.

Q16: What are the key differences between High-Level and Low-Level programming languages? (3 Marks)

- Low-level languages (Machine and Assembly) are machine-dependent and specific to one type of CPU.
- Machine language uses 0s and 1s, which are difficult for humans to write but directly understood by computers.

- High-level languages (like Python or Java) use English-like sentences and are machine-independent.
- High-level languages are easier for humans to code in but require translators to be understood by the computer.
- Assembly language uses mnemonics instead of binary but is still hardware-specific.

Q17: Explain the concept of Free and Open Source Software (FOSS). (3 Marks)

- FOSS refers to software where both the program and its source code are provided freely to the public.
- It allows users to study, improve, and add new functionality to the software.
- Ubuntu, Python, and LibreOffice are prominent examples of FOSS.
- It differs from 'freeware,' where the software is free to use but the source code is kept secret.
- Proprietary software, unlike FOSS, requires a purchase and its code is strictly copyrighted.

Q18: Describe the stages of data processing: Capturing, Storage, and Retrieval. (3 Marks)

- Data Capturing involves gathering raw data from digital sources like keyboards, barcode readers, or sensors.
- Data Storage is the process of saving captured data on digital devices (HDD, SSD) for future use.
- Data Retrieval involves fetching stored data from memory or a database to perform actions on it.
- Minimizing retrieval time is crucial for faster system performance.
- As data grows, storage and retrieval become more challenging tasks.

Q19: What is the role of language translators? Name three types. (3 Marks)

- Translators convert programs written in high-level or assembly language (source code) into machine code (object code).
- An Assembler converts assembly language mnemonics into machine code for a specific CPU.

- A Compiler translates the entire high-level source code into machine code at once.
- An Interpreter translates high-level code line-by-line during execution.
- Computers can only execute the resulting machine code, not the original source code.

Q20: Explain the need for secondary memory in a computer system. (3 Marks)

- Primary memory (RAM) is volatile and limited in storage capacity.
- Secondary memory is needed to store data and instructions permanently for future use.
- It is non-volatile, meaning it retains data even when power is off.
- It is generally cheaper and offers much larger storage capacity than primary memory.
- Examples include Hard Disks (HDD), Solid State Drives (SSD), and Pen Drives.

5 Mark Questions

Q21: Describe the components and functionality of the Von Neumann architecture. (5 Marks)

- The Von Neumann architecture is a foundational model for modern computer design.
- It consists of a Central Processing Unit (CPU) which contains the Arithmetic Logic Unit (ALU) and Control Unit (CU).
- The ALU performs calculations and logical operations, while the CU directs the flow of data and instructions.
- It includes a single memory space to store both program instructions and data.
- Input and Output devices allow the system to interact with the external world.
- Communication channels or buses are used to transfer data and signals between these components.
- The first binary programmable computer based on this architecture was the ENIAC.
- Refer to Figure 1.5 in the textbook for the block diagram of this architecture.

Q22: Detail the functions of an Operating System in managing a computer system. (5 Marks)

- Process Management: The OS manages multiple tasks (processes) in execution, allocating the CPU to finish tasks in minimum time.
- Memory Management: It dynamically allocates and frees main memory for running processes, ensuring they don't interfere with each other.
- File Management: The OS handles the creation, deletion, and protection of files in secondary storage, preventing unauthorized access.
- Device Management: It manages all connected I/O hardware using device drivers, ensuring heterogeneous devices work together.
- Security Management: The OS provides security via user authentication (passwords) and encryption to protect data from unauthorized users.
- Resource Allocation: It acts as a resource manager to ensure hardware components like RAM and CPU are used optimally.
- User Interface: It provides the platform (GUI, Command-based, etc.) through which users give commands to the hardware.

Q23: Explain the units of memory measurement and the hierarchy from Bit to Yottabyte. (5 Marks)

- The smallest unit of computer memory is a 'Bit', representing a 0 or a 1.
- A 'Nibble' consists of 4 bits, while a 'Byte' consists of 8 bits.
- Memory units increase by factors of 1024 (which is 2 to the power of 10).
- 1 Kilobyte (KB) = 1024 Bytes; 1 Megabyte (MB) = 1024 KB.
- 1 Gigabyte (GB) = 1024 MB; 1 Terabyte (TB) = 1024 GB.
- The hierarchy continues with Petabyte (PB), Exabyte (EB), and Zettabyte (ZB).
- The largest unit mentioned in the NCERT curriculum is the Yottabyte (YB), which equals 1024 ZB.
- Understanding these units is crucial for measuring data storage in modern devices like SSDs and cloud servers.

Q24: Discuss the evolution of computing devices from the Abacus to modern computers. (5 Marks)

- Abacus (approx. 500 BC): A mechanical device for simple arithmetic, considered the origin of computing.
- Pascaline (1642): Blaise Pascal's mechanical calculator for addition, subtraction, and repeated multiplication.
- Analytical Engine (1834): Charles Babbage's mechanical computing device which forms the basis of modern computers.
- Tabulating Machine (1890): Herman Hollerith's machine used punched cards to summarize data, a first step toward programming.
- Turing Machine (1937): Alan Turing's concept of a general-purpose programmable machine capable of solving any problem.
- EDVAC/ENIAC (1945): The introduction of stored-program computers where both data and programs reside in memory.
- Transistors (1947): Vacuum tubes were replaced by semiconductor transistors, making computers smaller and more efficient.
- Integrated Circuits (1970): Silicon chips containing entire electronic circuits drastically reduced computer size.
- Modern Era: Characterized by personal computers (1981), Laptops, Smartphones, and AI-driven IoT devices.

Q25: Explain the concept of Data Transfer between the CPU and Memory using the System Bus. (5 Marks)

- Data must constantly move between the CPU and memory for processing to occur.
- The System Bus, consisting of Data, Address, and Control buses, serves as the physical pathway.
- For a Read operation, the CPU places the target memory address on the Address Bus.
- The CPU then sends a 'read' signal through the Control Bus.
- A memory controller fetches the data from the specified address and places it on the Data Bus for the CPU.
- For a Write operation, the CPU provides the address on the Address Bus and the actual data on the Data Bus.

- A 'write' control signal is sent, and the data is recorded into the specified memory location.
- This complex coordination ensures that data is processed correctly and results are stored back.

Q26: Compare and contrast System Software and Application Software with examples. (5 Marks)

- System Software provides the basic functionality to operate hardware directly.
- Application Software is designed to cater to specific user requirements and works on top of system software.
- A computer system cannot function without system software (like an OS), but it can work without application software.
- System software manages resources (CPU, RAM), whereas application software performs user tasks (writing documents, editing photos).
- Examples of system software include Windows, Linux, Android, and device drivers.
- General-purpose application software includes Microsoft Word, VLC Player, and web browsers.
- Customized application software includes school management systems or banking websites tailored for specific organizations.
- Refer to Figure 1.13 for the categorization of these software types.

Q27: Discuss the different types of User Interfaces provided by modern Operating Systems. (5 Marks)

- Command-based Interface: Users type specific text commands (e.g., MS-DOS, Unix). It is keyboard-driven and less interactive.
- Graphical User Interface (GUI): Uses icons, menus, and windows. Interaction is primarily via mouse and keyboard (e.g., Windows, Ubuntu).
- Touch-based Interface: Used in smartphones and tablets, where touch gestures like scrolling and tapping are interpreted as commands.
- Voice-based Interface: Allows interaction via spoken commands, helpful for hands-free use or visually impaired users (e.g., Siri, Cortana, Google Now).

- Gesture-based Interface: Users interact via motions like waving, tilting, or eye movement (common in gaming and high-end smartphones).
- Each interface serves different accessibility needs and use cases.
- Refer to Figure 1.15 for a visual summary of OS user interface types.

Q28: Detail the security concerns related to digital data deletion and recovery. (5 Marks)

- Data deletion is one of the biggest threats, caused by hardware failure, accidental erasing, or malware.
- Standard deletion only marks a file's address as 'free' without actually wiping the bit-level data to save time.
- Data recovery is possible as long as the 'free' space hasn't been overwritten by new data.
- Security Concern 1: Unauthorized deletion by malicious software or users, which can be mitigated by using passwords and access controls.
- Security Concern 2: Unwanted recovery of sensitive data from discarded hardware.
- Mischief-mongers can recover 'deleted' files from old or broken storage devices, posing a threat to confidentiality.
- Mitigation: Users should use proper data shredding tools to permanently erase files before disposing of devices.
- Encryption is also a vital tool for protecting files from unwanted modification or reading.

Q29: Explain the role of Microprocessors and their key specifications. (5 Marks)

- A microprocessor is a CPU implemented on a single microchip.
- It is built using integrated circuits containing millions of transistors and other components.
- Word Size: The maximum number of bits a processor can process at once (currently ranges from 16 to 64 bits).
- Memory Size: The amount of RAM the processor can address, which increases with word size (up to 16 Exabytes for 64-bit).
- Clock Speed: Measured in Gigahertz (GHz), it is the number of pulses generated per second, indicating instruction execution speed.

- Cores: A core is a basic computation unit. Multicore processors (dual, quad, octa-core) can execute multiple tasks simultaneously.
- Modern microprocessors can process millions of instructions per millisecond.
- Refer to Table 1.2 for the evolution of microprocessor generations.

Q30: Explain the concept of Program Development Tools (IDEs, Editors, Debuggers). (5 Marks)

- Program development requires a suite of tools to write, translate, and test code.
- Text Editor: A software used to create text files where programmers type their instructions (source code).
- Translator: A required tool (Compiler or Interpreter) to convert the high-level source code into machine-readable object code.
- Debugger: A specialized software used to detect and correct errors (bugs) in the source code.
- Integrated Development Environment (IDE): A software that bundles an editor, building tools, and a debugger into one platform.
- IDEs allow programs to be typed, compiled, and debugged directly from a single interface.
- Examples include Python IDLE, Netbeans, Eclipse, and Atom.
- IDEs significantly simplify the software development life cycle for programmers.